Anthony Ibarra | Electrical & Computer Engineer

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Professional Profile

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective field. Showcasing a strong focus in Embedded Systems, Control System Theory, Robotics and Automation. With fundamental understanding of Pattern Recognition, Signal Processing, and Power Electronics making me a flexible Embedded & Control Systems Engineer. Possessing various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness for a great addition toward a multidisciplined team.

Core Hard SKILLS

- Embedded C/C++
- Python
- Bash
- MIPS Assembly
- FPGA, Verilog, VHDL
- BLE, UDP, TCP/IP, I2C,UART, SPI, Ethernet, RF, LTE
- HIL, Unit Testing, pytest

Core Soft SKILLS

- Soldering
- Testing
- Troubleshooting
- Problem Solving
- Creativity
- Designing
- Planning
- Leadership
- Motivation
- Management

Education & Qualifications

University of Illinois at Chicago UIC, Chicago IL — Jan 2023 - May 2025

Masters of Science in Electrical & Computer Engineering, ECE

Relevant Coursework:

- Audio & Acoustic signal processing
- Intro to filter Synthesis
- Adaptive Digital Filters
- Electromechanical energy Conversion
- Linear Systems Theory & Design
- Electromagnetic Compatibility
- Advanced Computer Communication
- Convex Optimization
- Mechatronics Embedded Design
- Intro Neural Networks

University of Illinois at Chicago UIC, Chicago IL — Aug 2017 - May 2022

Bachelors of Science in Computer Engineering, ECE

Relevant Coursework:

- Artificial Intelligence I
- Data Structures
- Foundation of Computing
- Principles of Modern Control
- Principles of Auto Control
- Robotics: Algorithm & Control
- Embedded Systems
- Computer Organization
- Advanced Computer Architecture
- Intro VLSI design
- Pattern Recognition I
- Computer Comm Networks I
- Probability and Random Process for engineers

Projects

Audio & Acoustic Signal Processing — Jan 2025 - May 2025

Research Paper(project)

Implement a solution for motorcycle riders with some degree of hearing impairment. The starting goal is to remove the dampening from the helmet, providing clear sound while wearing the helmet without adding any unneeded noise or unwanted sounds. In turn this can potentially amplify user awareness while riding or in largely populated areas. Report being in the scope and style of a Signal Processing Society conference paper.

- Develop a new method or solution with concepts to be applied to an existing problem. Perform real world testing and validation of project and research implementation
- Test and use skills and algorithms from acoustic & audio signal processing to achieve the set goal

Electromechanical Energy Conversion — Aug 2024 - Dec 2024

Series Brush DC Motor Modeling

From a given Datasheet of a motor replicate the graphs of speed, current, power, and efficiency vs. load torque. Use Ltspice to model the motor AMETEK Haydon Kerk Pittman Brush DC Motor 14207S007-SP, in an electrical circuit and simulate to obtain the graphs.

- Implement a behavior model of a DC motor using knowledge magnetic circuits and transformers; DC machines; and three-phase AC synchronous and induction machines
- DM – 100A DC Machine 0.3 kW (0.4 hp), 125 V, 2.4 A, and 1800 RPM; motor testing and verification, working with motors of up to 120v using 0 – 125 volt variable and 0 – 150 volt DC voltmeter /ammeters
- AC 120VA 60Hz transformer testing through AC measurements using wattmeters, voltage meters, and ammeters
- LTSpice design and simulation of DC motor mechanical behavior through electrical simulation

Electromagnetic Compatibility Project — Jan 2024 - May 2024

Optimization on Three Coil Long Range, Wireless Power Transfer

Based on the provided research papers attempt to recreate the results from said paper. Using Matlab, and HFSS. Provide a professional typeset report regarding the results with Latex.

- Electromagnetic Compatibility testing using a spectrum analyzer, Oscilloscopes for EMC and EMI testing for PCB and signal integrity using RF SMA coaxial cables, SMA terminators
- Create a three coil configuration for long range wireless power transfer for simulation, confirming data from research papers through simulations
- Mathematically simulated plot from Matlab regarding the Three coils as shown within the research paper
- Simulation and plot from Ansys HFSS 3D EM simulation software, simulating the three coil behavior in different configurations
- Make an IEEE formatted report using Latex, in a professionally typeset manner demonstrating discoveries and results

Mechatronics Embedded Design Project Jan 2023 - May 2023

Self Driving Car

Lead a team of 4 to design and develop an RC car to follow a line on a track. The car includes a DC motor for the four wheel drive, Servo for steering, and a Line camera for track detection.

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary
- Develop and tune controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

Intro to Artificial Neural Networks Jan 2023 - May 2023

Wildfire Classification

Design a neural-network-based wildfire/no wildfire classification, using a programming language (Python, Matlab, C++, etc.), and any framework(Pytorch, TensorFlow, Keras, Caffee, Matlab neural network toolbox, etc.). Dataset containing 10k images provided, the training and testing are split 9:1 respectively.

- Design a CNN (convolutional neural network) for pattern recognition and classification of real life cases
- Implement classification training for wildfires as it is similarly done for facial recognition
- Configure and tune the training and testing for the cnn model using python and pytorch with cuda cores and GPU's when able
- Use the created image classification model for wildfire prevention in testing

Robotics: Algorithm/Control Jan 2022 - May 2022

Mobile Robot Control

Program a Raspberry Pi alpha bot to complete tasks within a respective area. Programming language of choice is Python, with the robot containing many actuators and sensors.

- Embedded software and firmware development for the robot task completion
- Design kinematics for robot control and awareness for motion planning
- Implement video monitoring through onboard actuated camera
- Path awareness through implementation of arrays of infrared sensors; including robot localization through AruCo markers located around the environment
- Accomplish specific tasks within a constrained location

UIC Senior Design Aug 2021 - May 2022

Automated Watering System

Work in a four-student team to prototype a device that water specified plants by taking moisture levels, outdoor weather conditions, and plant information into account.

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device

Jan 2022 - May 2022 Pattern Recognition I

Face recognition, Principal Component Analysis (PCA)

Using Python scikit-learn datasets and matplotlib train and test, for face recognition.

- Using a large dataset from MNIST, adapt the dataset for training and testing scenarios
- Tune and adjust training phase accordingly while taking the training and testing results into account

Volunteer

Aug 2017 - May 2025 Table Tennis Club member TTC at UIC recreation center

Assist with opening, closing, and setting up equipment for club activities. Help with practice and training for future tournaments or events when possible.

2014 - 2017 Volunteer (volunteered 3 times)

HighSchool Heroes Program

The process of teaching 1st-3rd graders about society through a specified curriculum composed into four sections. Volunteers are placed in a group of 3 to complete said curriculum in a concise time frame for each section.

About Me

Main Focus:

- Embedded System
- Controls Systems
- Robotics/Automation

Interests:

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Neural Networks
- Programming
- Robotics
- Automation
- Research
- Development
- Design
- Embedded & Control systems
- Power Efficiency
- Cost Efficiency
- Hardware Security

Hardware used:

Equipment:

- Oscilloscope
- Multimeter
- Function Generator
- Waveform Generator
- PowerSupply
- Soldering Tools
- Breadboard
- Microcontrollers
- Keysight N9912A(FieldFox RF analyzer)

MCU:

- Tiva C launchpad TM4C123G (MCU: TM4C132GH6PM cortex M4)
- Arduino Nano 33 IoT (MCU: SAMD21 Cortex M0+)
- Raspberry pi (alpha bot) (MCU: Broadcom BCM2837B0, Cortex-A53)
- Freescale FRDM-KL25Z (MCU: Kinetis® L Series KL1x (KL14/15) & KL25 Cortex®-M0+)

Programming Languages:

- Python
- C/C++
- Assembly
- Verilog
- VHDL
- PLC(ladder logic)
- HMI (SCADA)
- CSS
- Html
- JavaScript
- Rust

Software used:

Programming:

- VS Code
- GitHub
- Git
- Ubuntu(WSL)
- Docker
- Spyder
- Microsoft Word, PowerPoint, Excel

Embedded:

- Code-Composer studio
- Kinetis design studio
- Arduino IDE
- Mars 4_5
- VNC viewer
- STM32 CubeMX
- DMK Extraction tool
- ControlFlash

Debuggers:

- GNU debugger
- LLDB
- OpenOCD
- CMSIS
- Sterllaris ICDI
- U-Link

Robotics

- ROS2

Circuit design:

- Altium Designer
- Quartus
- Flux

- LTspice

Simulation & Design or Development:

- MATLAB
- Solidworks
- Mathematica
- Design Spark
- Blender
- HFSS Ansys

PLC-HMI-SCADA:

- RSLogix Micro
- RSLinx Classic
- RSLogix Emulate 500
- Connected Component Workbench (CCW)
- CODESYS
- Ignition

Database:

- mySQL
- SQL

Test Automation:

- Robot Framework
- Selenium

Hobbies:

- Repairing (disassembling and reassembling electronic parts / devices)
- Table-Tennis, video games, DIY projects